

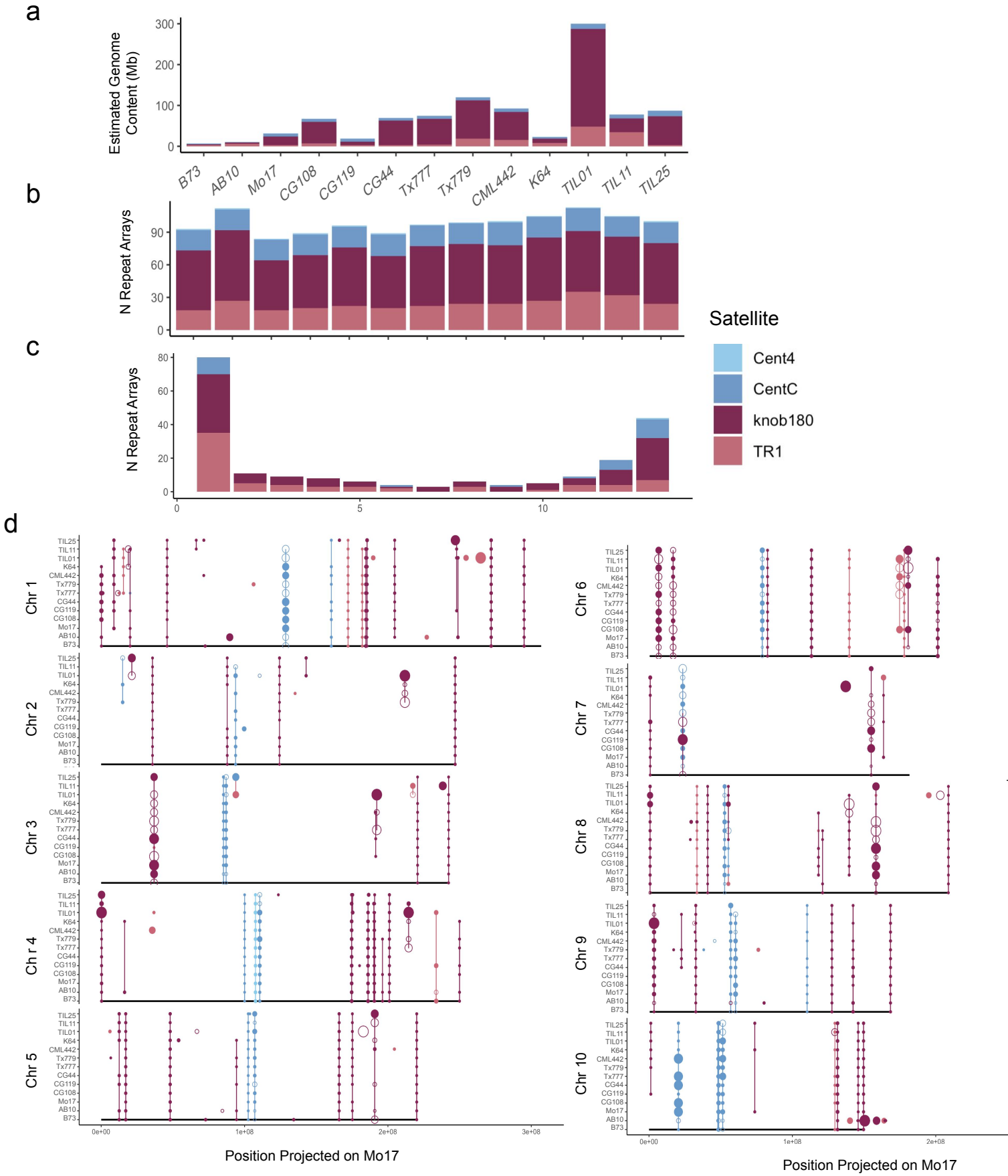


Figure 1. Assembly Repeat Content. a) Total repeat content by satellite. b) Repeat array counts. c) Frequency spectrum of arrays. d) Satellite arrays projected onto the T2T Mo17 assembly, the reference for scaffolding. Open circles represent arrays that are not fully assembled (contain an N gap). Lines

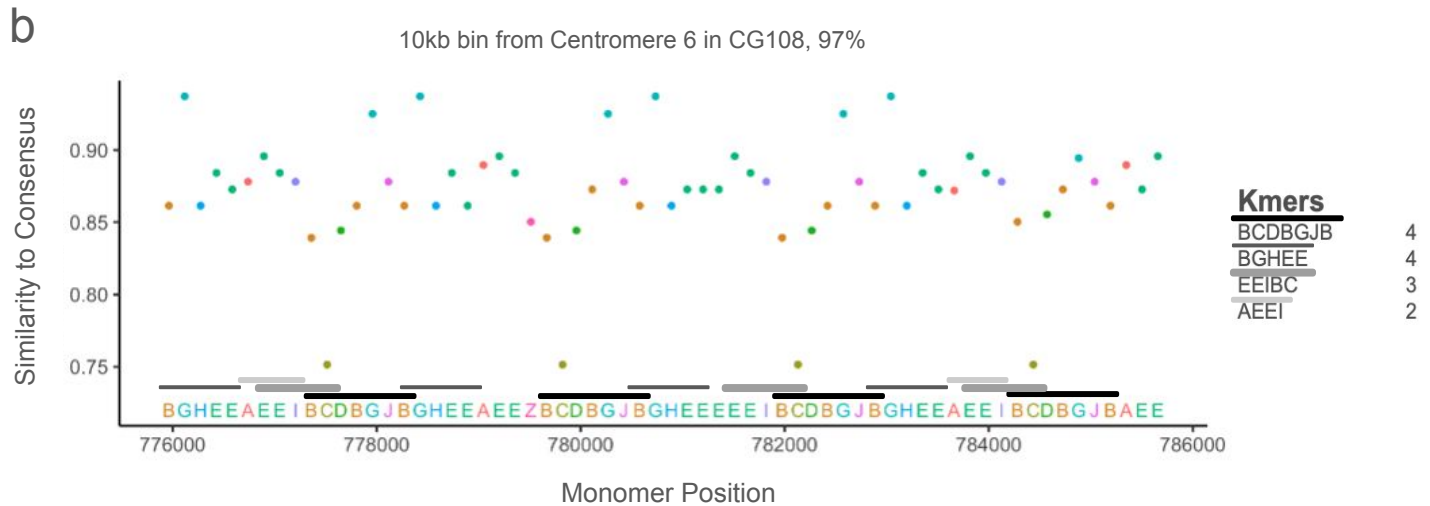
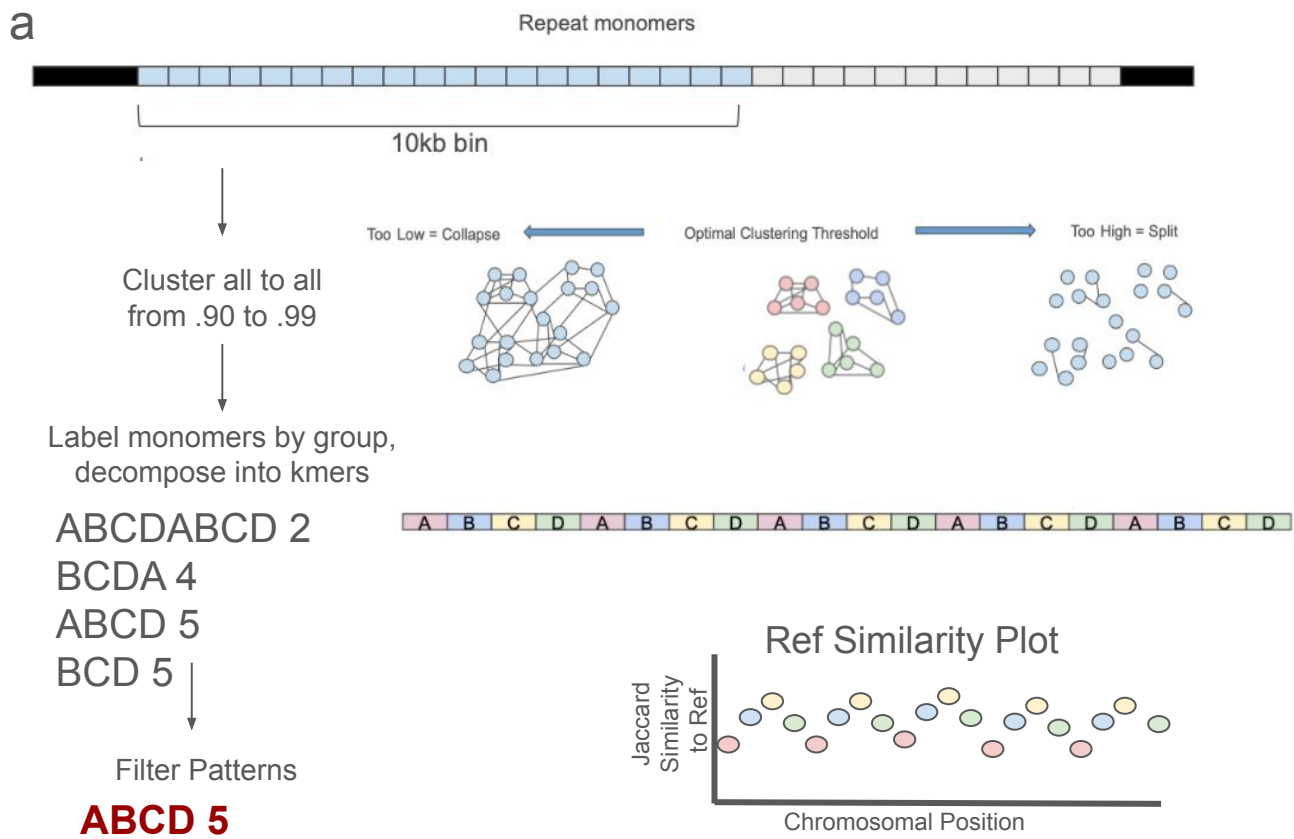


Figure 2. Local HOR Identification. a) Monomers in 10kb bins are extracted from the genome and compared all-to-all using BLAT. Jaccard similarity scores are calculated. Then, similarity networks are generated for thresholds from .99 to .90, in increments of .01. Each node represents a monomer sequence. If two monomers are identical, they are collapsed into one node. If two monomer sequences are at least as similar as the threshold, monomers are connected with an edge. Each distinct cluster in the network is labeled with a letter. Private clusters, or clusters that only contain a single, uncollapsed monomer, are labeled with “Z”. Monomers, with labels of their cluster identity, are then put back in their original genomic order, resulting in a character string. The character string is then decomposed into k-mers, starting with k=3 until all kmers have a frequency of 1. Kmers are then filtered – larger kmers that occur less frequently than its subset kmers, smaller kmers that occur as frequently as larger kmers containing it, and kmer that contain the private monomer “Z”. Patterns can be further assessed using a reference similarity plot, where the y axis represented Jaccard similarity to consensus, and x axis represents genomic position. Each dot represents a monomer. b) Sample HOR region, where monomers are labeled and colored by cluster identity. HOR kmers and their frequencies are listed on the side, and each repeated pattern in the arrays is marked with a line corresponding to its kmer.

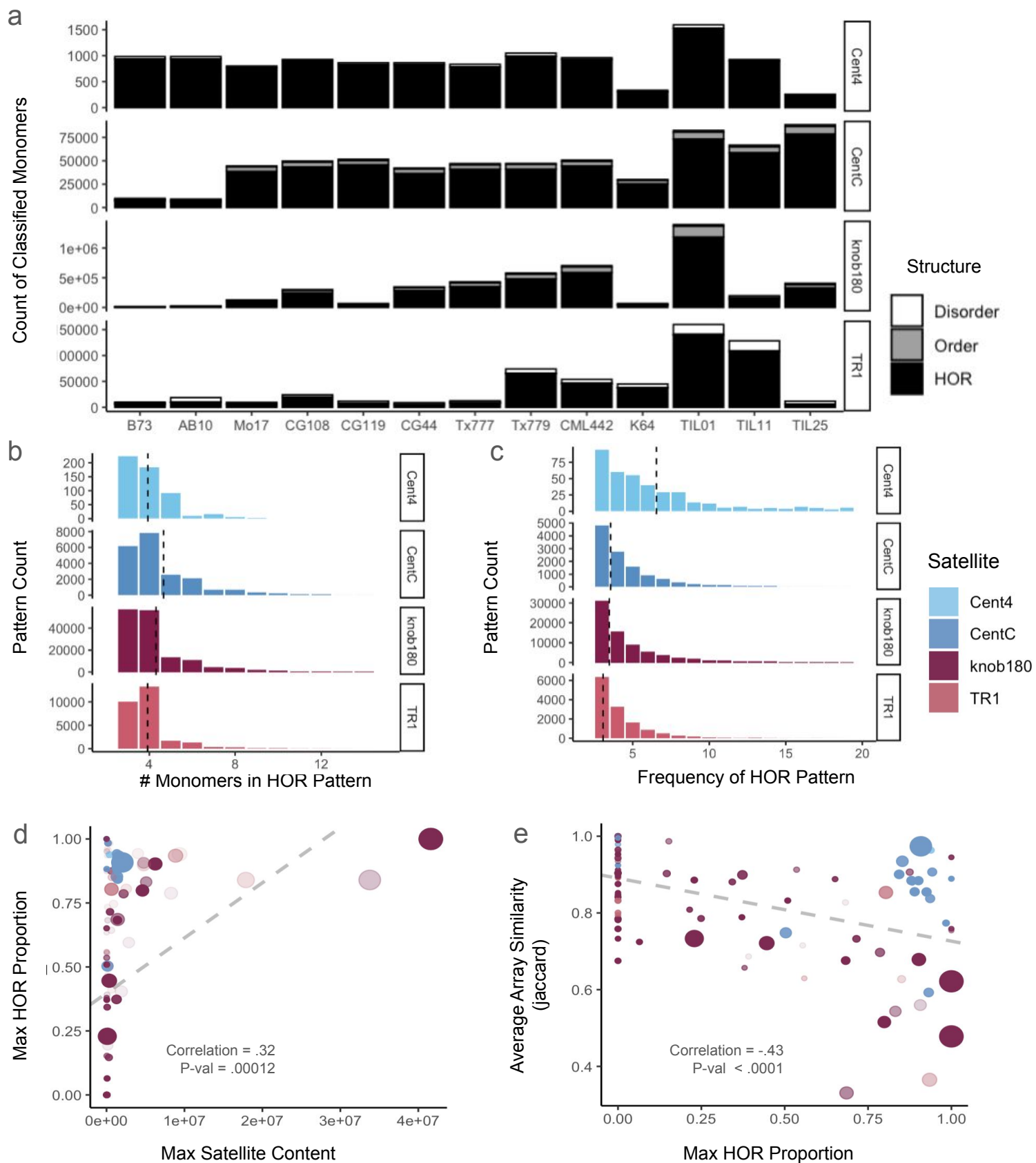
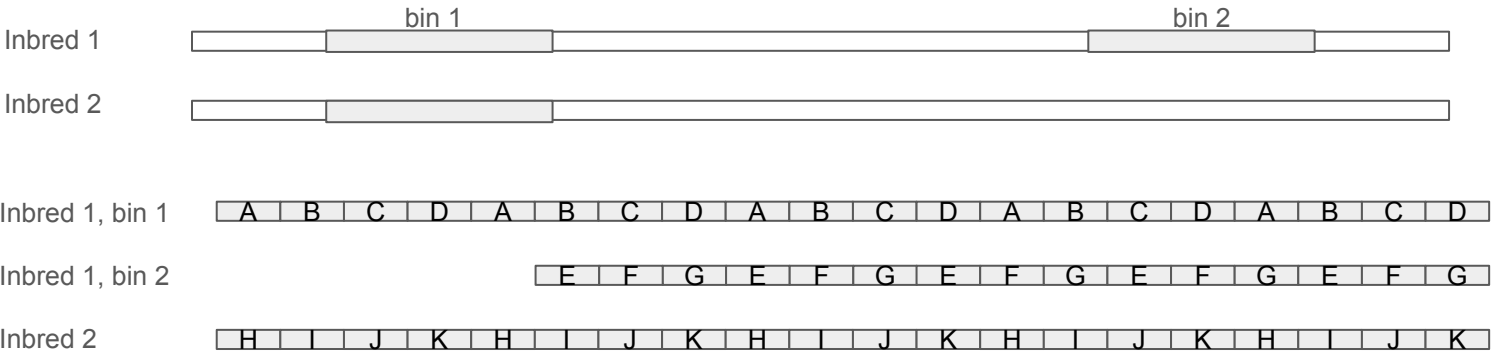
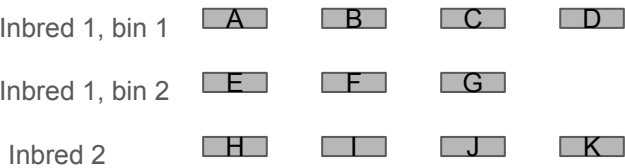


Figure 3. HOR Pattern Frequency and Length. a) HOR pattern length distribution. b) HOR pattern frequency distribution. Dashed line marks mean. c) Relationship between maximum pattern length within an inbred by satellite and total satellite content. Dashed line marks mean. d) Within a homologous group of arrays across the 13 genomes, represented by a single dot, maximum satellite content of a single array

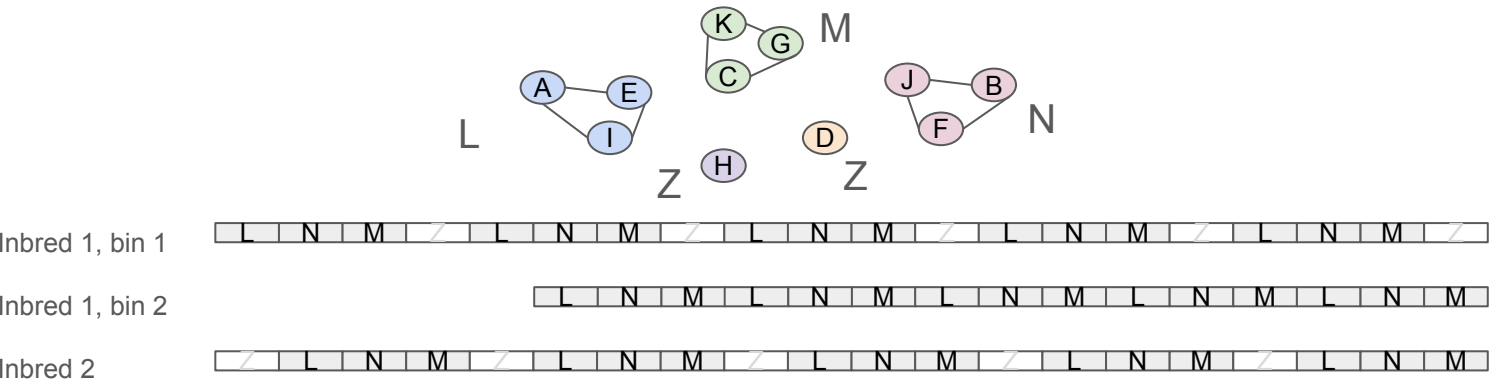
1. Collect HOR regions with the same optimal clustering threshold from a single homologous position



2. Generate consensus sequences for all monomers in local HOR patterns



3. Cluster consensus monomers and relabel, private monomers labeled “Z”



LN M is a shared HOR pattern

4. Recalculate Purity

Region	Local Purity	Shared Purity
Inbred 1, bin 1	1.00	.75
Inbred 1, bin 2	1.00	1.00
Inbred 2	1.00	.75

Figure 4. Shared HOR Identification. 1. Within a homologous group of arrays across the 13 genomes, all regions with confirmed HORs with the same optimal clustering thresholds (i.e. purity of HOR content identified in process outlined in Figure 3.1 is >0) are extracted. This can include multiple distinct regions from the same array 2. Consensus sequences for each monomer subtype in an original HOR pattern 3

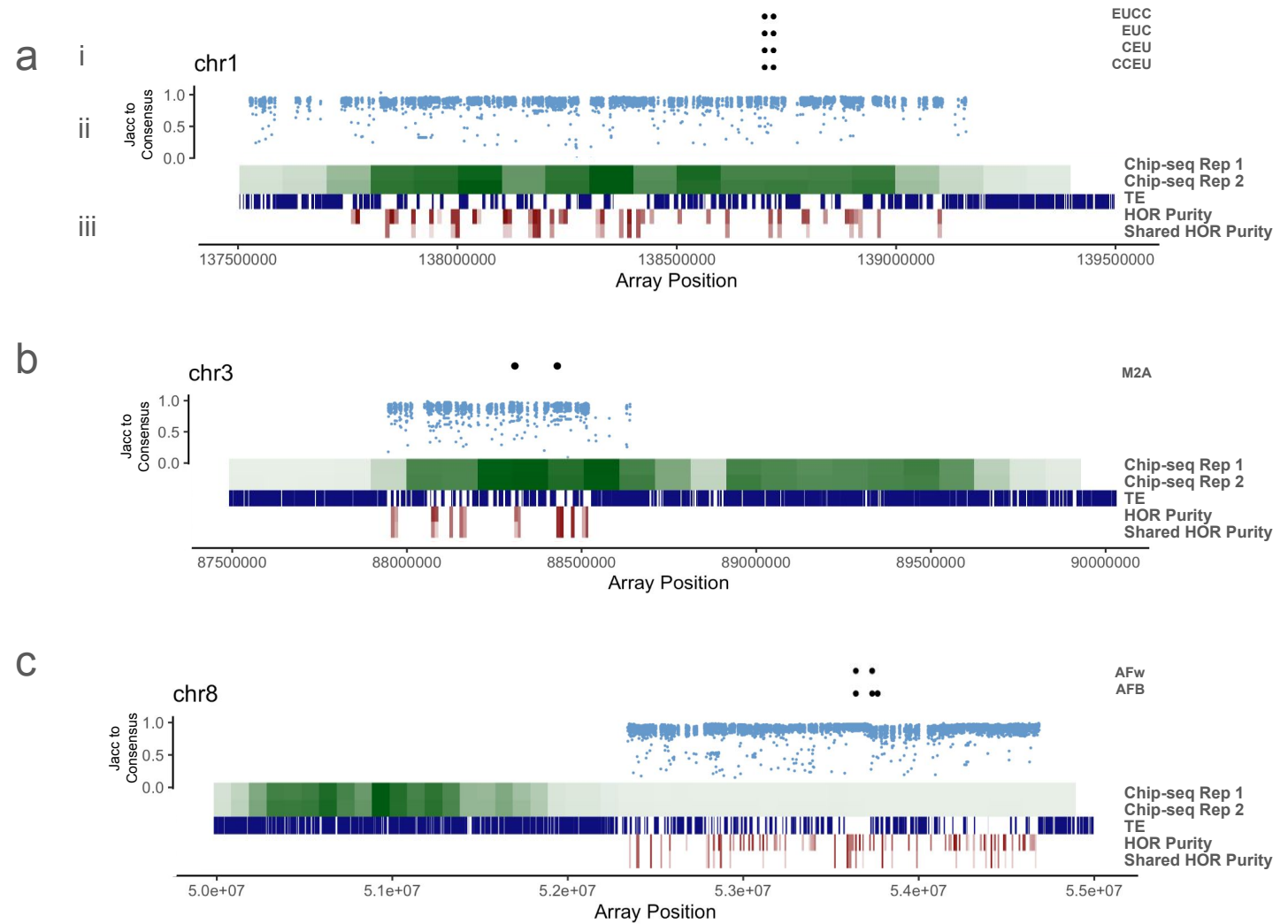
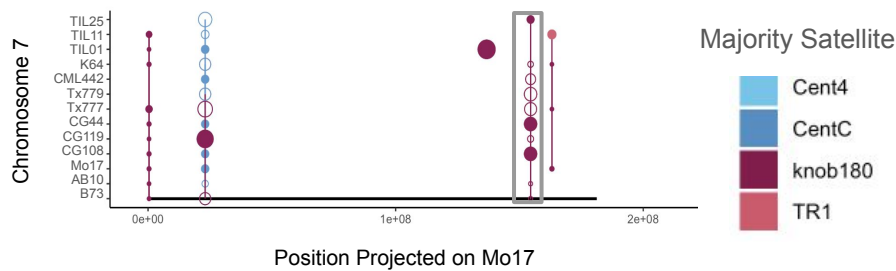
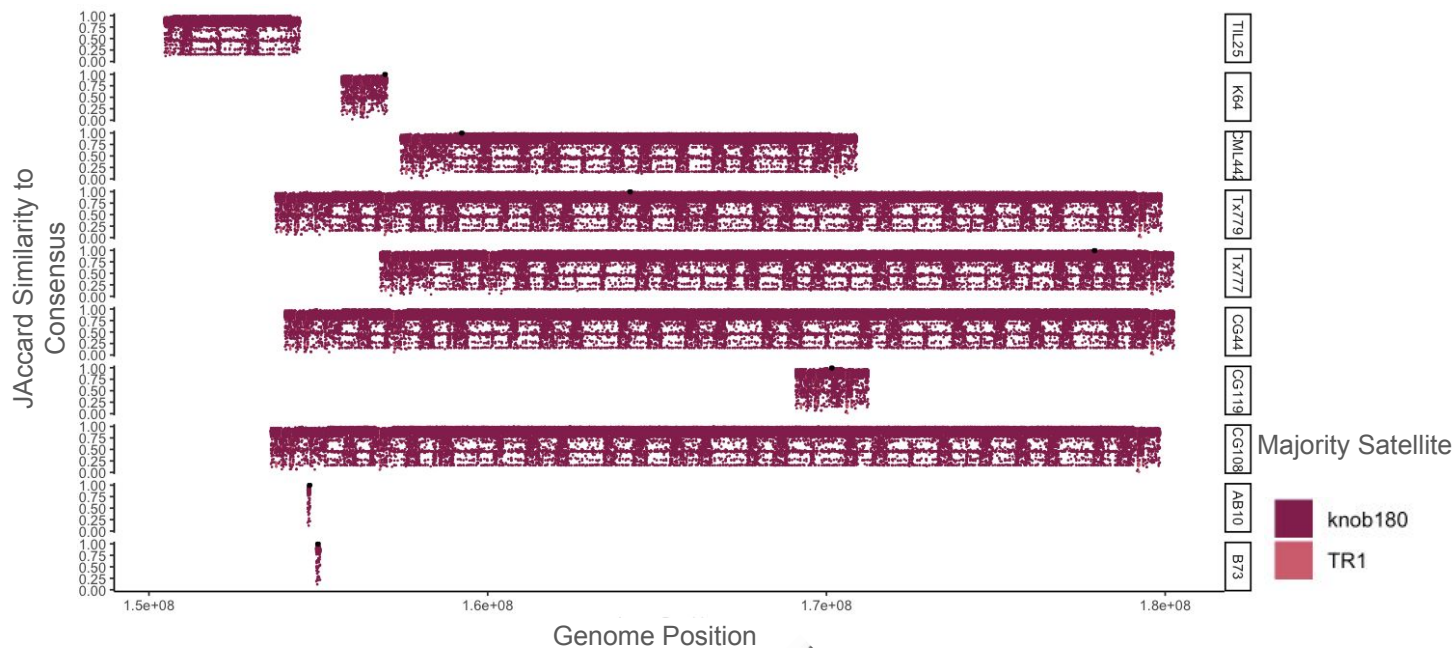


Figure 5. CENH3 ChIP-seq on Mo17. a) Centromere structure of chromosome 1. i) i) Shared HOR patterns that occur in at least two distinct HOR regions. Dot indicates presence in array, x axis is genomic position. ii) Similarity dot plot of all CentC monomers to consensus. iii) Relative ChIP-seq density in 100kb bins represented by shades of green. TE presence is represented by blue boxes. HOR purity and shared HOR purity are represented in red in 10kb bins. Darker shades of red represent higher purity values. b) Centromere structure of chromosome 3. c) Centromere structure of chromosome 8.

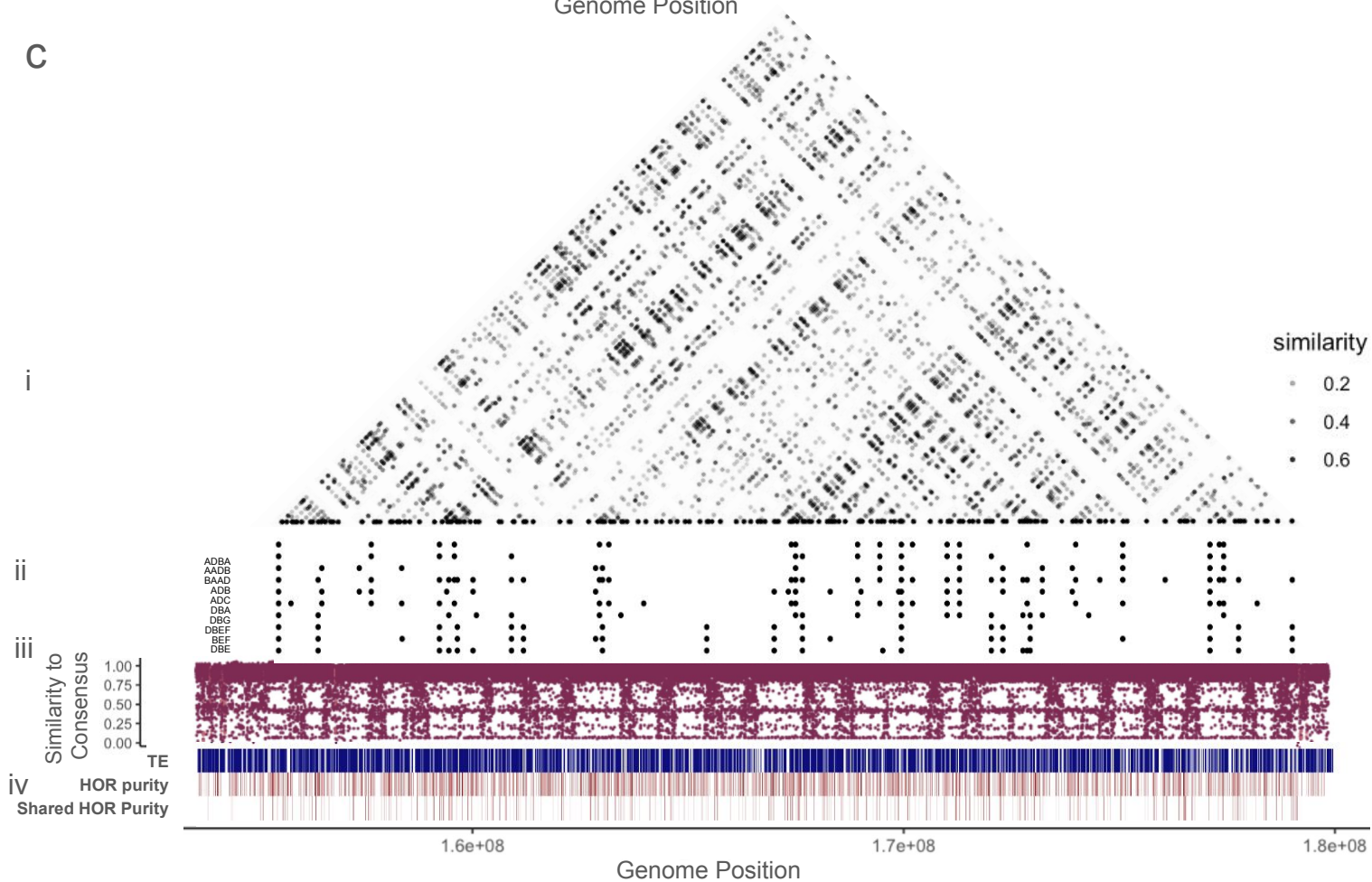
a



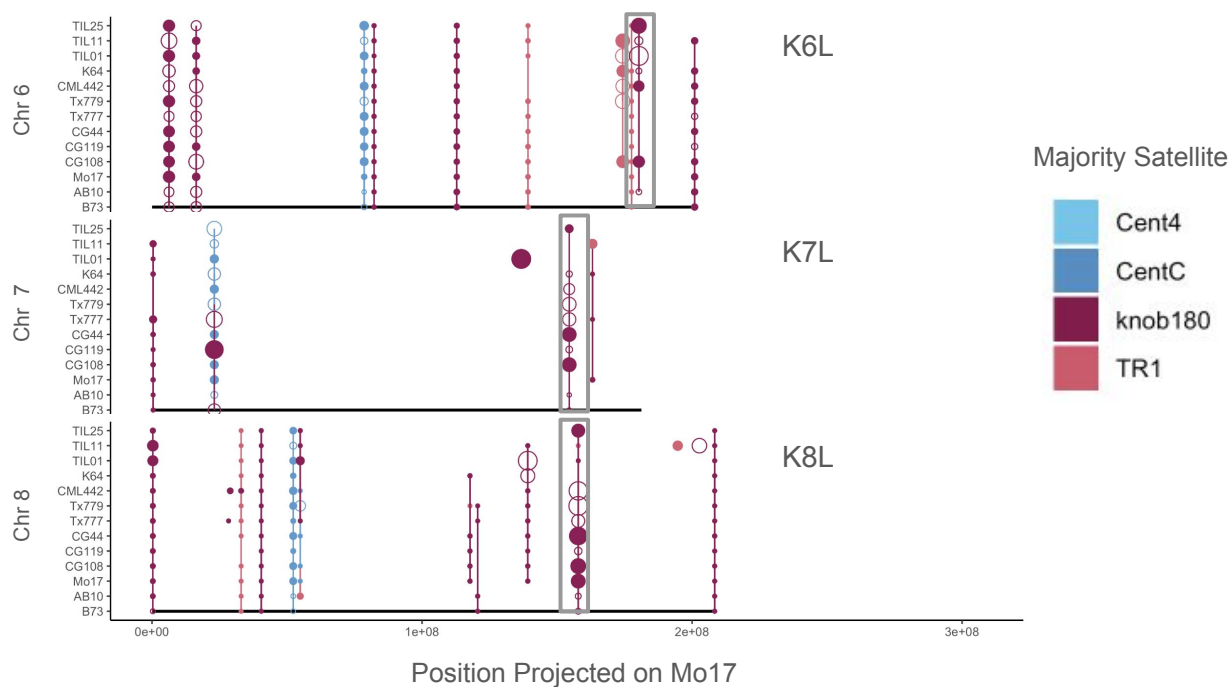
b



c



a



b

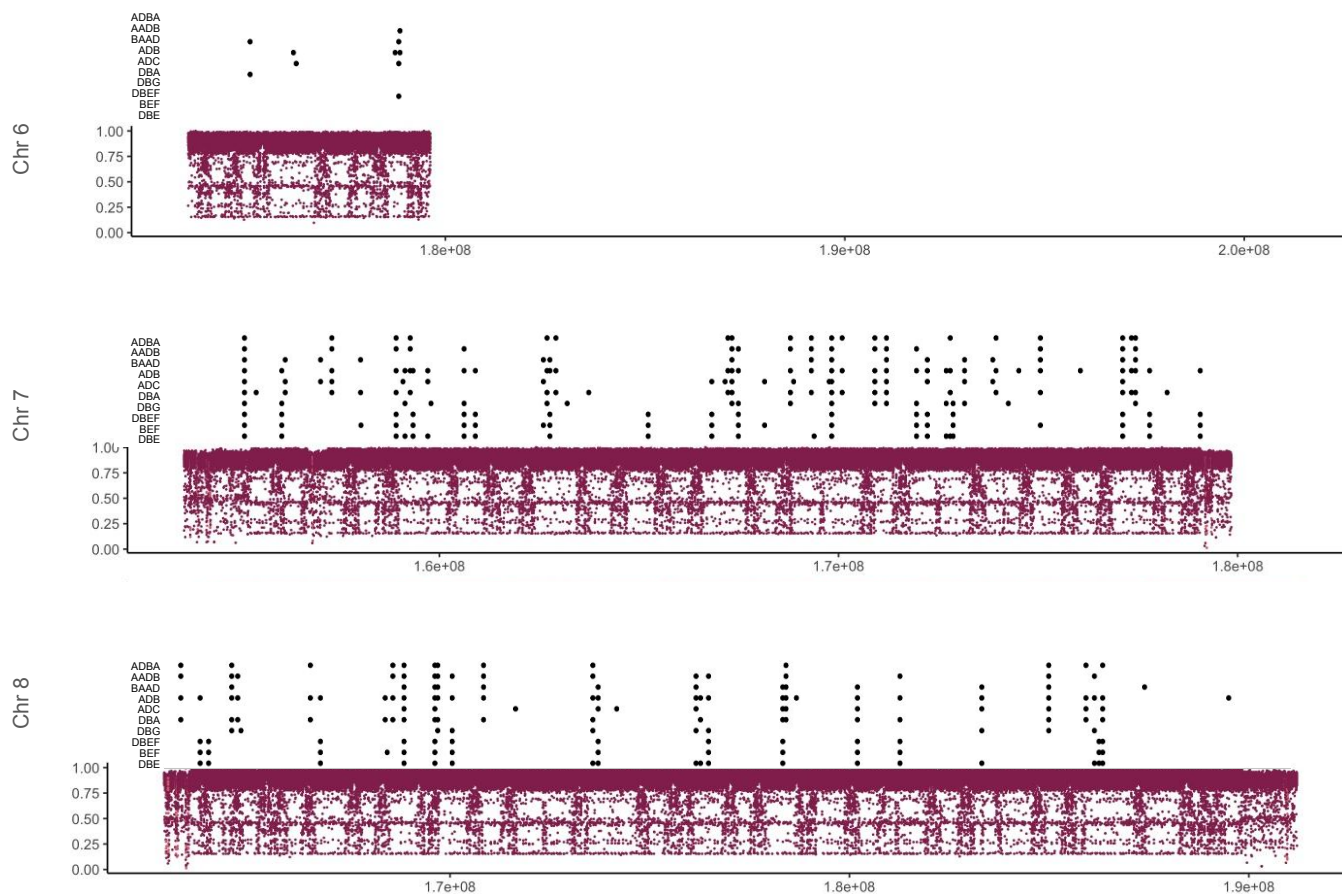


Figure 7. Shared HORs Among Knob180 Arrays. a) K6L, K7L, and K8L presence in the 13 genomes, as marked by the grey boxes. Fully assembled knobs are represented by filled circles. b) High-frequency HOR patterns. Dot indicates presence in array, x axis is genomic position.

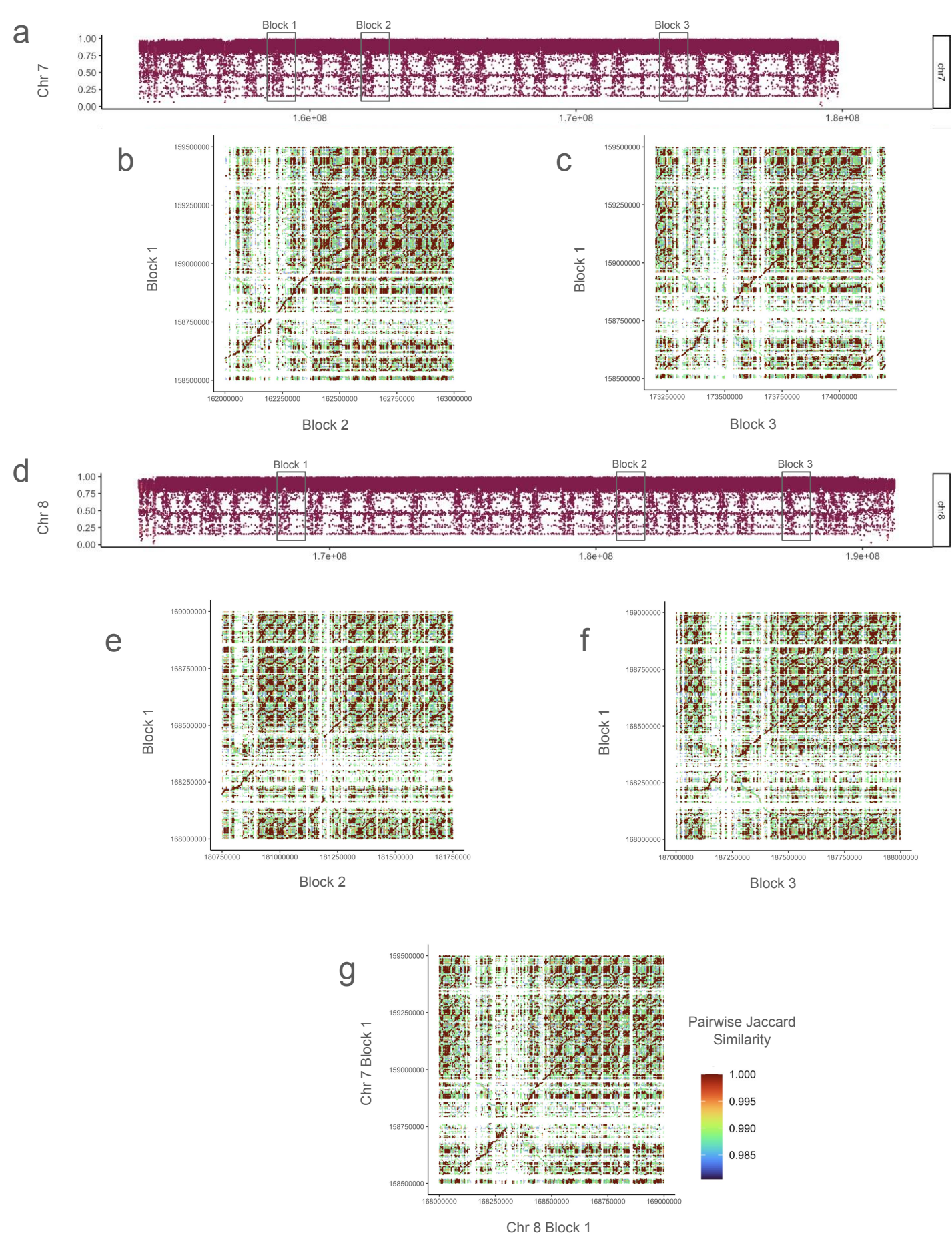


Figure 8. Similarity Blocks. a) CG108 K7L monomer dot plot to consensus. b) All to all monomer

Supplement

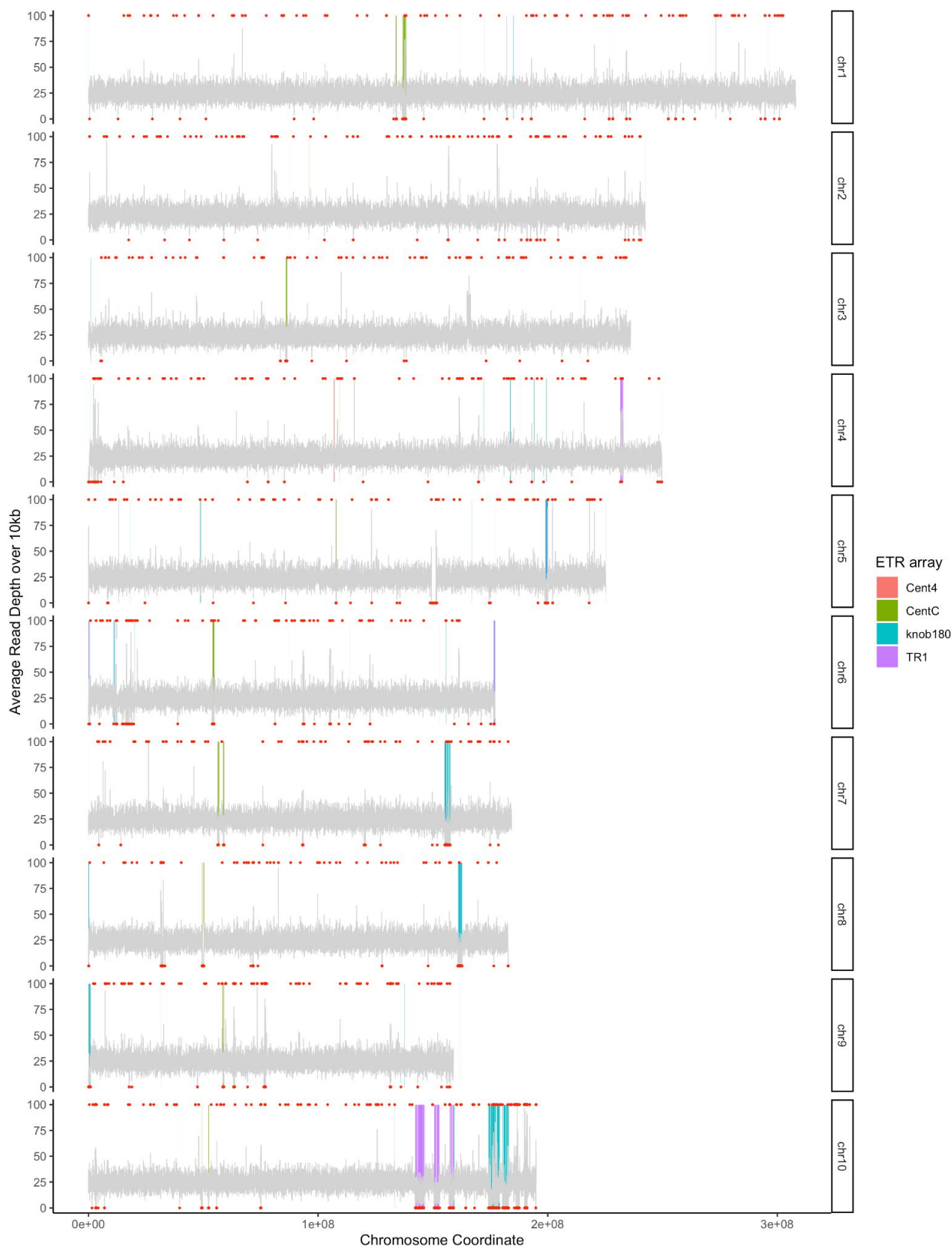


Figure S1. PacBio HiFi reads aligned to Gapless AB10 Pseudomolecules. Grey lines represent 10kb average read depth. Red dots above and below make 10kb bins where the average read depth falls outside two standard deviations of the mean depth. Satellite arrays are represented by colored boxes.

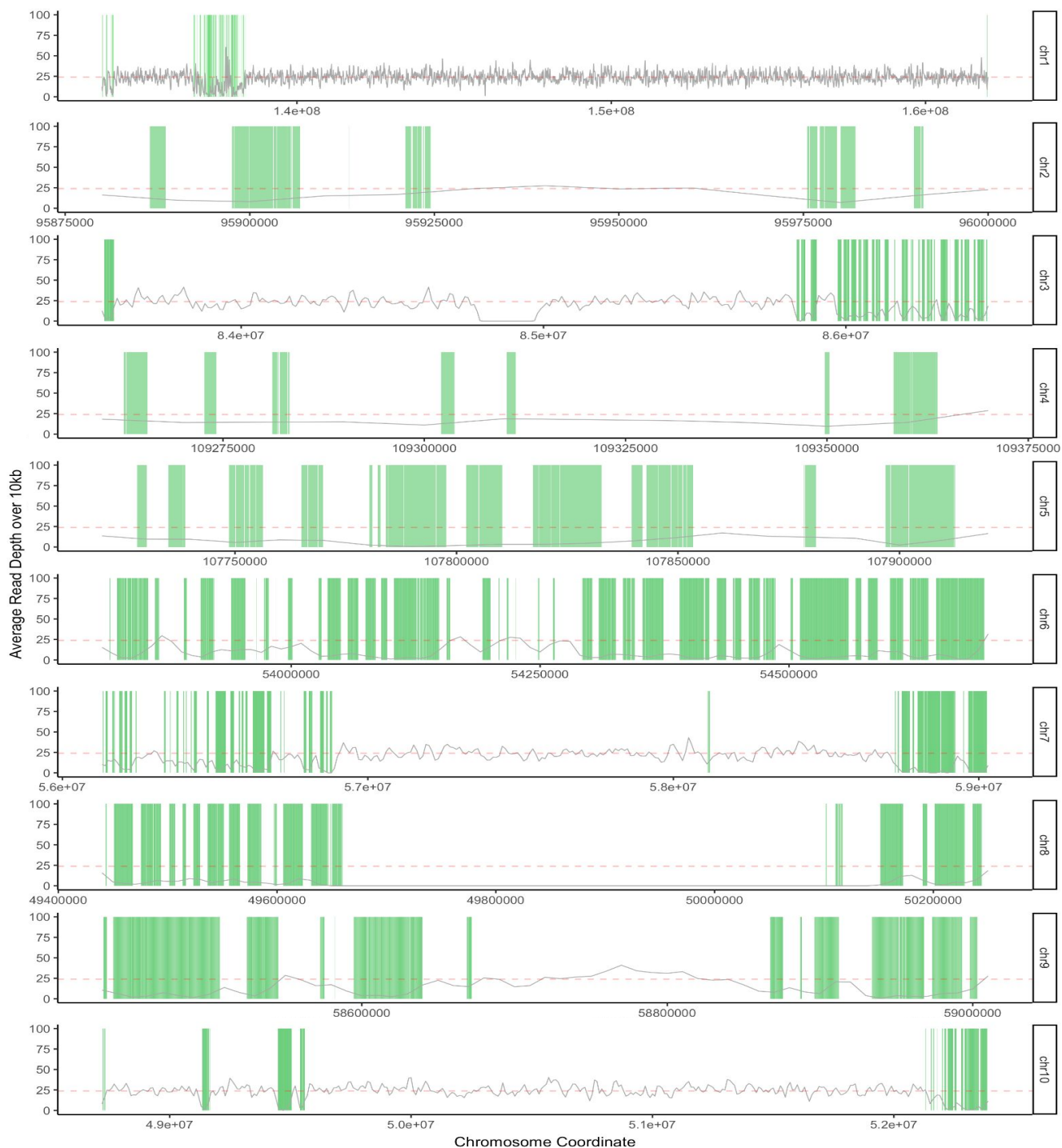
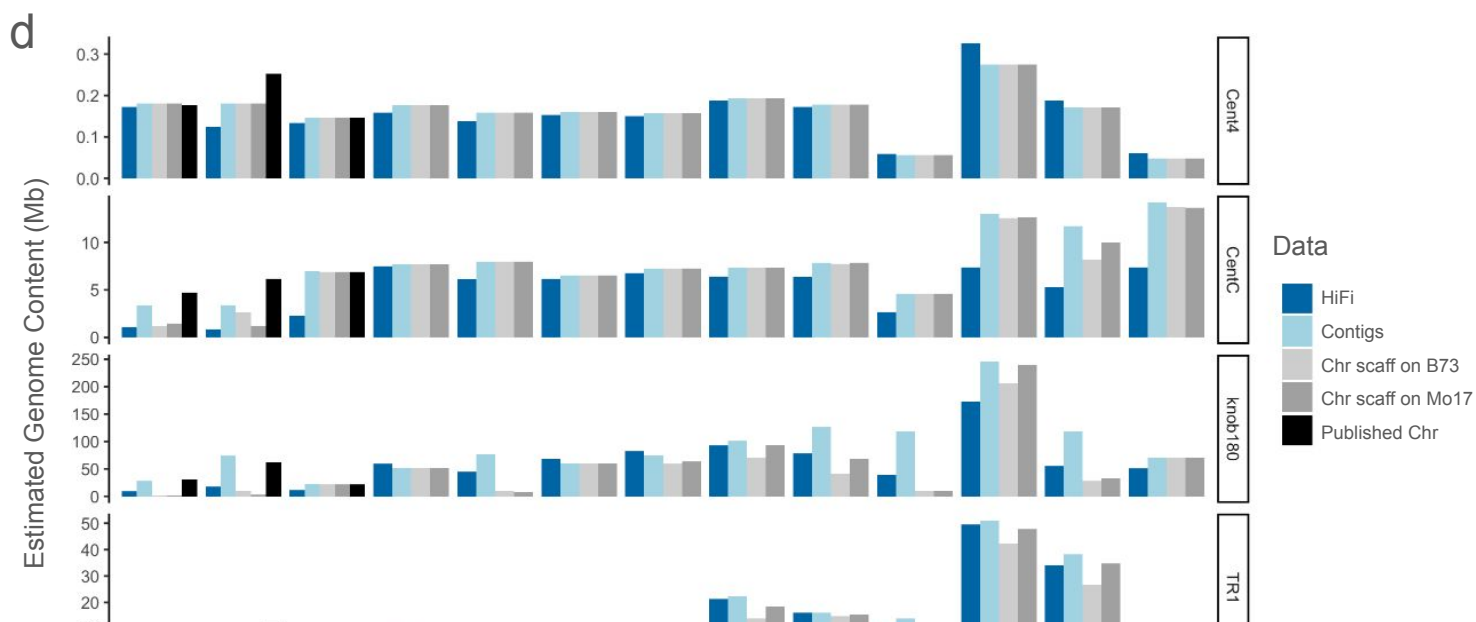
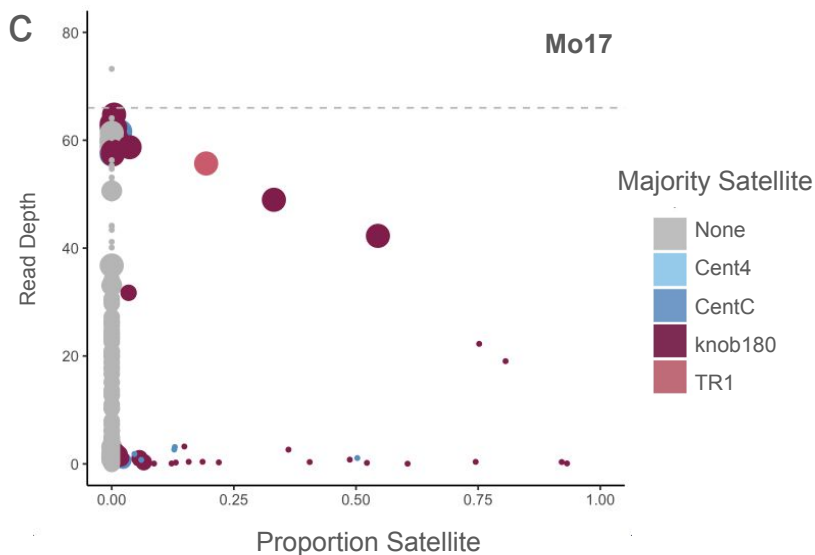
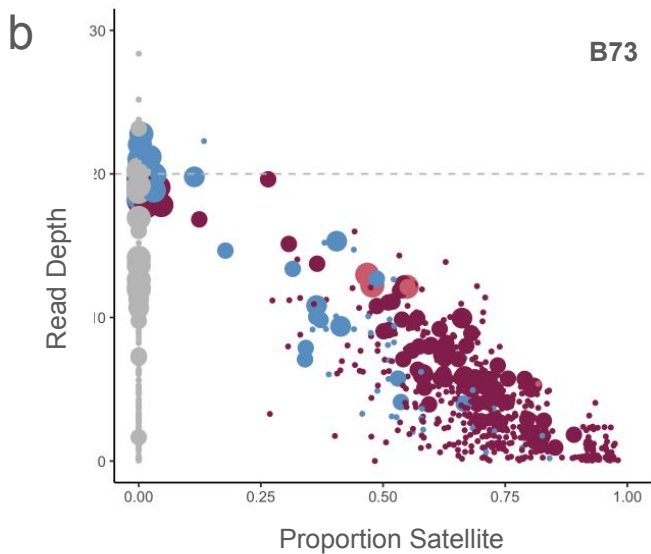
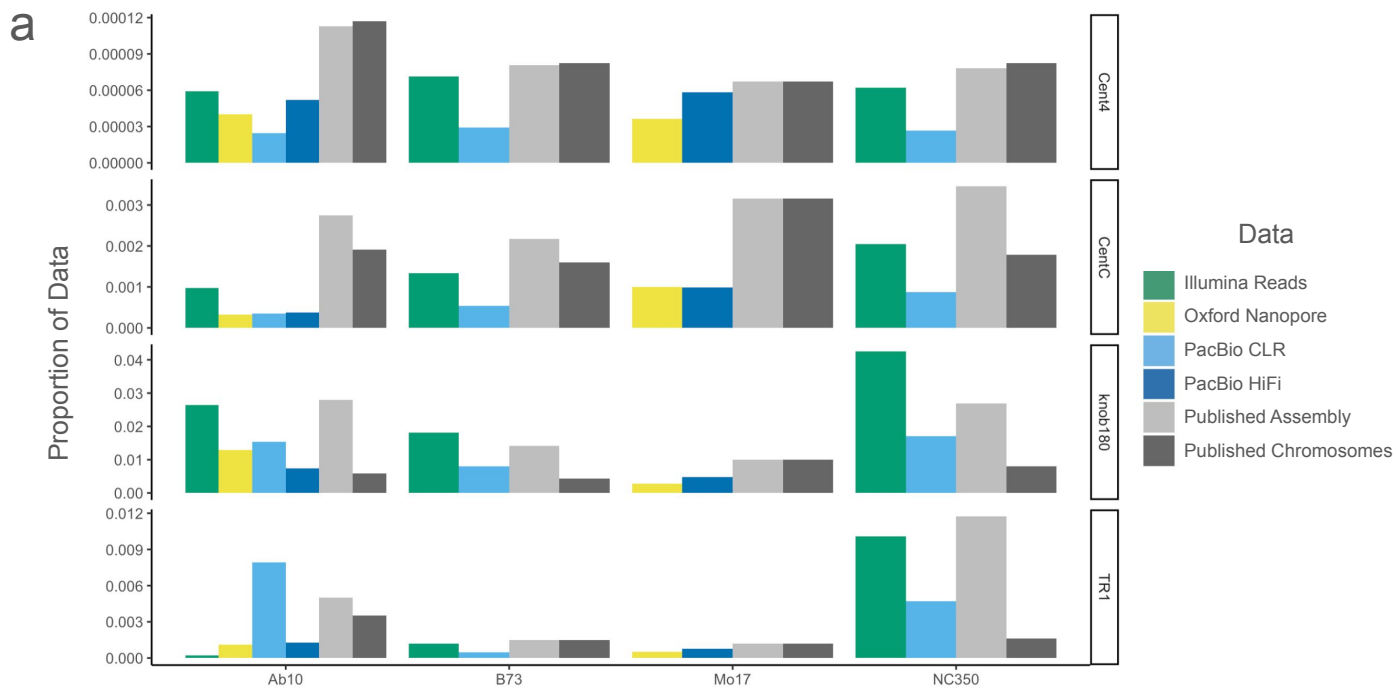


Figure S2. PacBio HiFi reads aligned to Gapless AB10 Centromeres. Grey lines represent 10kb average read depth. Red dotted line represents the average read depth. CentC satellites are represented by green boxes.



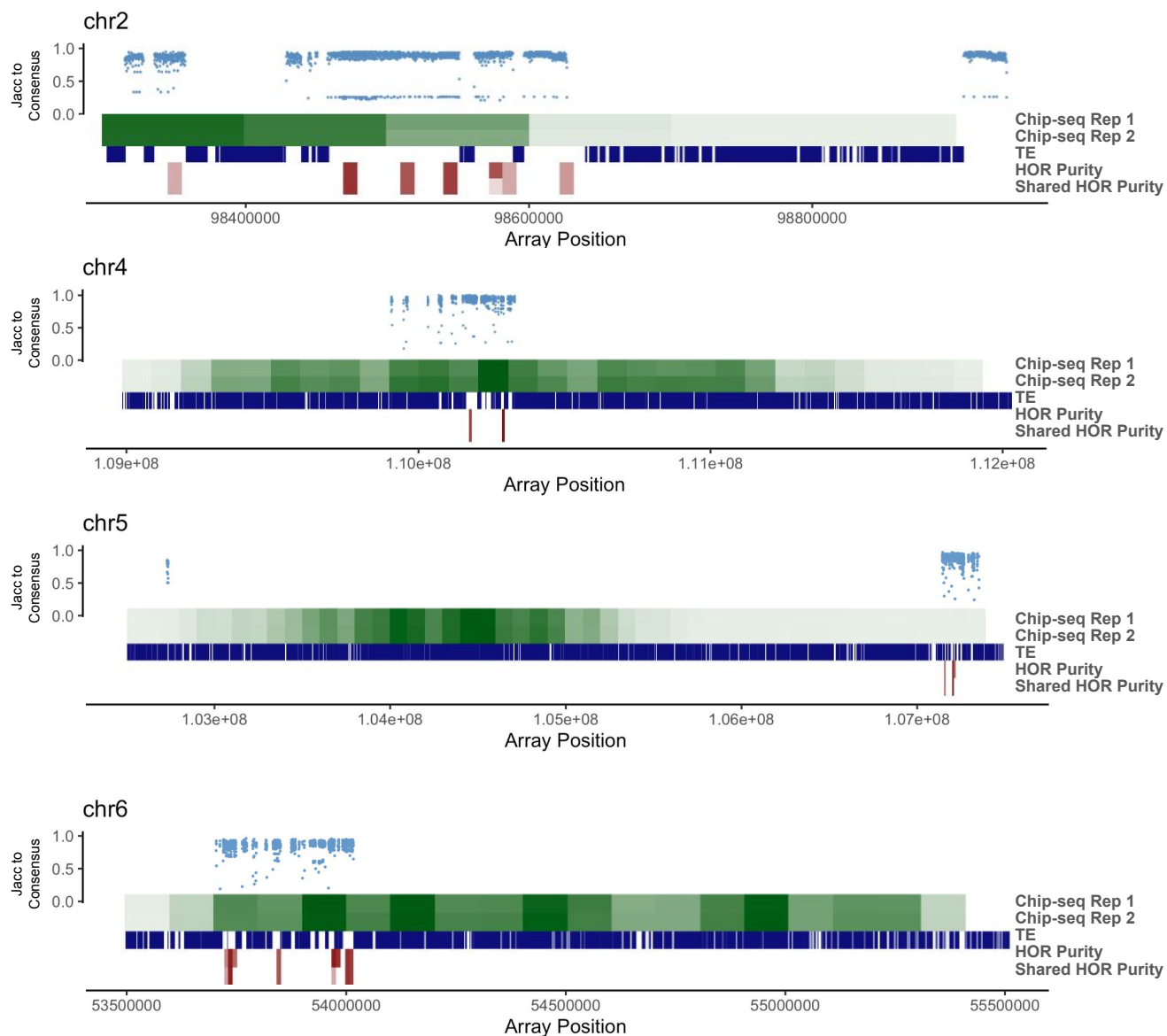


Figure S4. CENH3 ChIP-seq on Mo17. Centromere structure of chromosomes 2, 4, 5, and 6. Relative Chip-seq density in 100kb bins represented by shades of green. TE presence is represented by blue boxes. HOR purity and shared HOR purity are represented in red in 10kb bins. Darker shades of red represent higher purity values.

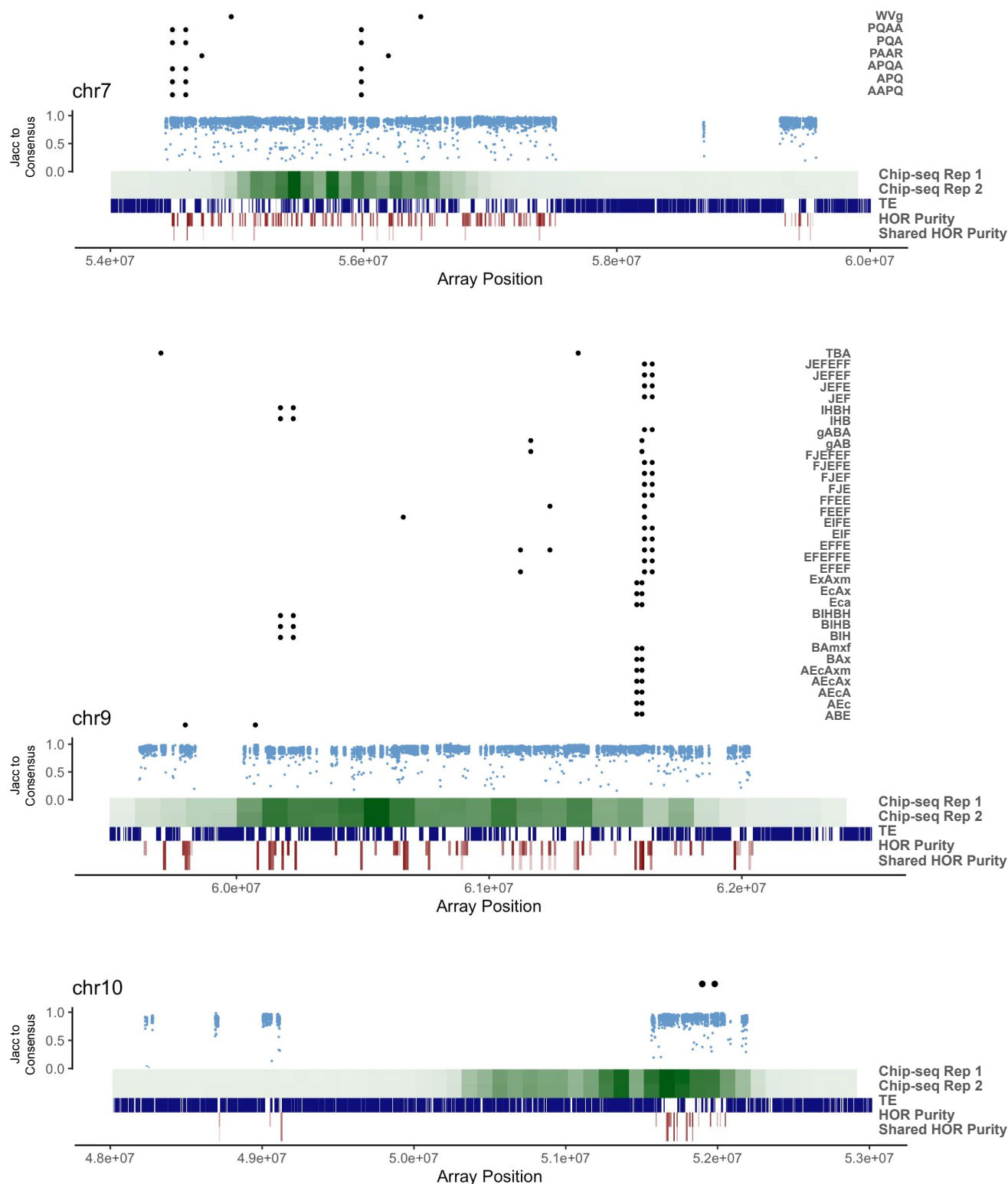


Figure S5. CENH3 ChIP-seq on Mo17. Centromere structure of chromosomes 7, 9, and 10. Shared HOR patterns that occur in at least two distinct HOR regions. Dot indicates presence in array, x axis is genomic position. Relative ChIP-seq density in 100kb bins represented by shades of green. TE presence is represented by blue boxes. HOR purity and shared HOR purity are represented in red in 10kb bins. Darker shades of red represent higher purity values.